

Project 1: Instructions: Retail Sales Performance Analysis (Power BI Desktop)

Project Overview

You've joined the **Sales Analytics** team at a retail company. Senior management wants a Power BI report and a short presentation to help them: monitor sales performance, identify trends, compare regions and product categories, and understand the impact of discounts and costs.

Scope for Project 1 (keep it simple):

- Work in **Power BI Desktop only**
- **No publishing required**
- **No data modelling / relationships required**
- **No mobile layout required**

1) Files Provided

- Dataset (CSV): **Retail Sales Performance Analysis Dataset.csv**
- Project guide reference: **Retail Sales Performance Analysis_Project_Guide.pdf**

2) Required Deliverable (Submit)

PowerPoint presentation only

Submit **one** file:

- **.pptx** (short and professional, with screenshots as evidence)

You may follow the original 12-slide structure if you want, but it's not mandatory.

Assessment focus (you are NOT assessed on visual perfection)

- Logical reasoning
- Quality of decisions
- Accuracy of calculations
- Clarity of communication
- Ability to translate data into business insight

3) Dataset Fields (what you're working with)

The dataset contains transaction-level retail sales records, including:

- Order information: OrderID, OrderDate, ShipMode
- Customer segmentation: CustomerSegment
- Geography: Country, State, City, Region
- Product: Category, SubCategory
- Numeric fields: Quantity, UnitPrice, Discount, Unit_Cost

4) Part 1 — Data Understanding & Preparation (Power Query)

Required (simple, but real):

1. Load the CSV into Power BI Desktop (Import).
2. Check / set data types (minimum expectation):
 - OrderDate → Date
 - Quantity → Whole number
 - UnitPrice, Discount, Unit_Cost → Decimal number
3. Identify at least **2 data issues** and decide how you will handle them (examples in the original guide include returns/zero values/unusual discounts). Document your decisions.

Note: The sample data shows OrderDate in a day/month/year style format (e.g., 31/07/2022). Make sure it becomes a real Date field.

PPT evidence required (screenshots)

Include screenshots of:

- Your **Power Query** steps pane showing key changes (data type fixes + issue handling)
- A short bullet list of the **2+ issues** you found and what you did

5) Part 2 — Core Calculations (DAX Measures)

Create the following **6 required measures** (explicit measures; do not rely only on drag-and-drop implicit totals).

These match the project guide's intent that revenue and profit must be calculated from components.

1. **Total Revenue**
 - Total Revenue = Quantity × UnitPrice × (1 – Discount)

2. **Total Cost**

- $\text{Total Cost} = \text{Quantity} \times \text{Unit_Cost}$

3. **Total Profit**

- $\text{Total Profit} = \text{Total Revenue} - \text{Total Cost}$

4. **Profit Margin %**

- $\text{Profit Margin \%} = \text{Total Profit} \div \text{Total Revenue}$
- If Total Revenue is 0, return **blank or 0** (state which you chose and why)

5. **Total Quantity Sold**

- Sum of Quantity

6. **Number of Orders**

- Distinct count of OrderID
- Note: counting rows may inflate “orders” if OrderID repeats across transaction lines

PPT evidence required (screenshots)

Include screenshots of:

- Your **Measures list** (Fields pane / Model pane) showing the six measure names
- One screenshot of a **card visual** (or table) that displays Total Revenue + Total Profit + Profit Margin % to prove measures work

6) Part 3 — Build the Senior Management Report Page (Power BI Desktop Only)

Goal

Create **one clear report page** in Power BI Desktop that a senior manager could use to understand performance quickly.

Important: This is a **report page**, not a Power BI Service “dashboard”.

Requirements (what must be on the page)

Your report must include one main page containing the following visuals (using your measures):

1. **KPI Cards** (minimum 3)
 - Total Revenue
 - Total Profit
 - Profit Margin %

2. Sales Trend Over Time

- A line chart (or area chart) showing performance over OrderDate

3. Performance by Category / SubCategory

- A bar or column chart showing either:
 - Profit by Category, or
 - Profit by SubCategory (Top N recommended), or
 - both (two visuals)

4. Regional Performance

- A bar chart OR map showing performance by Region or State

5. Top Performers

- A table or chart highlighting top performing SubCategories (or similar)

Page design rules

- Put KPIs at the top (first thing a manager sees).
- Each visual must have a clear title describing what it answers.
- Keep it readable: avoid overcrowding; whitespace is allowed.

PPT evidence required (screenshots)

Include **1 full-page screenshot** of your completed report page.

7) Part 4 — The “One Fine Challenging Task” (Required)

Challenge: Discount Impact on Profitability

Senior management explicitly wants to understand discount impact.

Create a visual (or small visual section) that answers:

“At what discount levels do we lose profitability?”

Requirements:

1. Create discount bands (manual bands recommended):
 - 0%
 - 1–10%
 - 11–20%
 - 21–30%
 - 31%+

2. Show at least:
 - Total Revenue
 - Total Profit
 - Profit Margin %
by discount band
3. Write **2–4 bullet insights** under the visual (in the report screenshot area or in the PPT):
 - Which discount band is most profitable?
 - Which band appears risky (margin drops or profit turns negative)?
 - Any category/region you would investigate further?

PPT evidence required (screenshots)

Include:

- Screenshot of the **discount band visual**
- Your **2–4 insights bullets**

8) PPT Structure (recommended, simple)

Your PPT can follow this structure:

1. Title + project brief
2. Dataset overview + key fields
3. Data issues found + how you handled them (with Power Query screenshot)
4. Measures created (definitions + evidence screenshot)
5. Senior management report page (full-page screenshot + 3–5 insights)
6. Discount challenge results (visual screenshot + recommendations)
7. Limitations + next steps

9) Submission Checklist (learner-facing)

- [] Loaded the CSV and corrected key data types (especially OrderDate).
- [] Documented at least 2 data quality issues + decisions (with evidence screenshot).
- [] Created 6 required measures using the project-standard definitions (with evidence screenshot).

- [] Built 1 main **report page** with required visuals (with full-page screenshot).
- [] Completed the discount-impact challenge visual + short insights (with screenshot).
- [] Submitted **one PPTX** containing all required screenshots and explanations.